Experiment 15: Refined OFDM Delay Estimation under Different SNR Conditions

Sionna RT-Based Multipath Channel at 5.9 GHz

V2X–ISAC Research

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Abstract

This experiment investigates the robustness of OFDM-based multipath delay estimation under different signal-to-noise ratio (SNR) conditions using a channel generated through NVIDIA Sionna RT.

A 5.9 GHz carrier frequency, 100 MHz sampling frequency, 1024-point FFT, and QPSK modulation are used. The ray-tracing environment produces twelve valid propagation ray entries, from which five prominent delay-domain reference peaks are identified.

Complex additive white Gaussian noise (AWGN) is introduced at the receiver for six SNR conditions ranging from -5 dB to 20 dB. Each condition is evaluated using 100 Monte Carlo trials.

The evaluation incorporates one-to-one peak matching, fractional-sample delay interpolation, detection probability, precision, recall, F1-score, false peaks per trial, delay mean absolute error (MAE), delay root mean square error (RMSE), and confidence intervals.

The results show that increasing SNR substantially improves detection precision and delay estimation accuracy. Precision increases from 35.08% at -5 dB to 100% at 20 dB, while delay RMSE decreases from 3.334 ns to 0.354 ns. The F1-score increases from 0.496 to 0.963.

The experiment also identifies a weaker multipath component whose detection remains challenging even at high SNR, highlighting the importance of detection threshold selection in multipath sensing applications.

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1 Introduction

In the previous experiments, OFDM channel estimation and delay-profile reconstruction were investigated under noise-free conditions.

Although these experiments established the numerical consistency of the channel estimation and delay-domain processing stages, practical wireless communication and sensing systems operate in the presence of noise.

The objective of Experiment 15 is therefore to evaluate the robustness of delay-domain peak detection under different SNR conditions.

The experiment uses the same Sionna RT-derived multipath channel as the previous experiments. AWGN is introduced at the receiver, and the resulting delay profiles are evaluated using an improved peak detection and matching methodology.

The main objectives are:

1. Investigate the influence of SNR on delay-profile estimation.
2. Evaluate the detection probability of dominant multipath components.
3. Quantify false detections and missed detections.
4. Improve delay estimation using fractional-sample interpolation.
5. Evaluate estimation accuracy using MAE, RMSE, and delay bias.
6. Investigate the detection behaviour of weaker multipath components at different SNR levels.

2 Simulation Configuration

The experiment uses the `simple_street_canyon` scene available in Sionna RT.

The transmitter and receiver positions remain unchanged from the previous experiments.

$$\mathbf{p}_{TX} = [-20, 0, 2]^T \text{ m} \tag{1}$$

$$\mathbf{p}_{RX} = [20, 0, 2]^T \text{ m} \tag{2}$$

The transmitter-receiver separation is:

$$d_{TX,RX} = 40 \text{ m} \tag{3}$$

The simulation parameters are presented in Table 1.

Table 1: Experiment 15 simulation parameters

Parameter	**Value**
Ray-tracing environment	Simple street canyon
Backend	LLVM
Carrier frequency	5.9 GHz
Sampling frequency	100 MHz
FFT size	1024
Modulation	QPSK
Cyclic prefix	128 samples
Subcarrier spacing	97.65625 kHz
Useful symbol duration	10.24 microseconds
Discrete channel length	60 samples
Valid ray entries	12
Noise model	Complex AWGN
SNR values	-5, 0, 5, 10, 15, 20 dB
Monte Carlo trials	100 per SNR
Detection threshold	-25 dB
Peak matching tolerance	10 ns

3 Ray-Tracing Channel and Reference Peaks

The Sionna RT simulation produced twelve valid propagation ray entries.

The earliest and latest propagation delays were:

$$\tau_{\min} = 133.426 \text{ ns} \tag{4}$$

$$\tau_{\max} = 283.282 \text{ ns} \tag{5}$$

The ray delays were converted into a discrete-time channel using fractional-delay windowed-sinc interpolation.

The noise-free channel response was subsequently transformed into the delay domain. Five prominent reference peaks were identified using a local maximum detector and a -25 dB threshold.

The reference peak positions after parabolic interpolation are presented in Table 2.

Table 2: Reference delay-domain peaks

Peak	**Reference delay (ns)**	**Reference magnitude**
1	131.150812	-4.271 dB
2	149.600922	0.000 dB
3	176.494451	-16.015 dB
4	230.029754	-24.807 dB
5	281.620815	-22.940 dB

These five reference peaks represent prominent delay-domain components of the sampled channel. They are not necessarily equivalent to five independently resolved physical

propagation rays.

4 Mathematical Methodology

4.1 OFDM Signal Model

The received OFDM signal is represented by:

$$y[n] = \sum_{m=0}^{N_h-1} h[m]x[n-m] + w[n] \tag{6}$$

where:

- $x[n]$ is the transmitted OFDM signal.
- $h[m]$ is the discrete multipath channel.
- $w[n]$ is complex AWGN.
- N_h is the discrete channel length.

The channel contains 60 samples, while the cyclic prefix contains 128 samples. Therefore, the cyclic prefix exceeds the channel memory in this static simulation.

The noise is introduced after ideal cyclic-prefix removal and synchronization.

4.2 AWGN Model

The complex AWGN is defined as:

$$w[n] = \sqrt{\frac{\sigma_w^2}{2}}\,(w_I[n] + jw_Q[n]) \tag{7}$$

where $w_I[n]$ and $w_Q[n]$ are independent standard normal random variables.

The noise variance is determined from the desired linear SNR:

$$\mathrm{SNR}_{\mathrm{linear}} = 10^{\frac{\mathrm{SNR}_{\mathrm{dB}}}{10}} \tag{8}$$

The noise variance is:

$$\boxed{\sigma_w^2 = \frac{P_y}{10^{\mathrm{SNR}_{\mathrm{dB}}/10}}} \tag{9}$$

where P_y is the average received signal power before noise addition.

The measured signal power was:

$$P_y = 1.258177 \times 10^{-11} \tag{10}$$

For example, at an SNR of 10 dB:

$$\sigma_w^2 = \frac{1.258177 \times 10^{-11}}{10^{10/10}} \tag{11}$$

$$= \frac{1.258177 \times 10^{-11}}{10} \tag{12}$$

$$= 1.258177 \times 10^{-12} \tag{13}$$

Similarly, at -5 dB:

$$\sigma_w^2 = \frac{1.258177 \times 10^{-11}}{10^{-5/10}} \tag{14}$$

$$\approx 3.978 \times 10^{-11} \tag{15}$$

The noise variance decreases as the SNR increases.

4.3 Frequency-Domain Channel Estimation

The transmitted QPSK symbols are known at the receiver. The estimated channel frequency response is calculated using:

$$\widehat{H}[k] = \frac{Y[k]}{X[k]} \tag{16}$$

where:

- $X[k]$ is the transmitted frequency-domain symbol.
- $Y[k]$ is the received frequency-domain symbol.
- $\widehat{H}[k]$ is the estimated channel response.

4.4 Delay-Domain Transformation

The estimated channel response is converted into the delay domain using the inverse discrete Fourier transform:

$$\widehat{h}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \widehat{H}[k] e^{j2\pi kn/N} \tag{17}$$

where $N = 1024$.
The sampling interval is:

$$T_s = \frac{1}{f_s} \tag{18}$$

For a sampling frequency of 100 MHz:

$$T_s = \frac{1}{100 \times 10^6} \tag{19}$$

$$= 10 \text{ ns} \tag{20}$$

Therefore, the discrete delay positions are separated by 10 ns before interpolation.

4.5 Fractional-Sample Peak Estimation

Integer-sample peak detection limits the estimated delay to discrete sampling positions.

To improve delay estimation, parabolic interpolation is applied around each detected local maximum.

Let y_{-1}, y_0, and y_{+1} represent the log-magnitude values at the left neighbouring sample, the detected peak, and the right neighbouring sample.

The fractional offset is:

$$\delta = \frac{1}{2} \frac{y_{-1} - y_{+1}}{y_{-1} - 2y_0 + y_{+1}} \tag{21}$$

The interpolated sample position is:

$$n_{\text{refined}} = n_{\text{peak}} + \delta \tag{22}$$

The estimated propagation delay is:

$$\boxed{\widehat{\tau} = \frac{n_{\text{peak}} + \delta}{f_s}} \tag{23}$$

The interpolation offset is restricted to the interval $[-0.5, 0.5]$ samples.

This method provides a sub-sample estimate of the location of a sampled peak. It does not independently increase the physical resolution of the waveform.

4.6 One-to-One Peak Matching

The detected peaks are compared with the five noise-free reference peaks.

A detected peak is considered a match when:

$$|\widehat{\tau}_j - \tau_i| \leq 10 \text{ ns} \tag{24}$$

where τ_i is a reference delay and $\widehat{\tau}_j$ is a detected delay.

Each reference peak and each detected peak can participate in at most one match.

This one-to-one assignment prevents a single detected peak from being counted as multiple successful detections.

5 Performance Metrics

5.1 True Positives, False Positives and False Negatives

The peak-matching procedure classifies detections into three categories:

- True Positive (TP): a detected peak matched to a reference peak.
- False Positive (FP): a detected peak that does not match any reference peak.
- False Negative (FN): a reference peak that was not detected.

For each SNR condition, the total number of reference peak opportunities is:

$$N_{\text{reference}} = N_{\text{trials}} \times N_{\text{peaks}} \tag{25}$$
$$= 100 \times 5 \tag{26}$$
$$= 500 \tag{27}$$

The following metrics are calculated from the aggregate counts across the 100 trials.

5.2 Detection Probability

Detection probability measures the proportion of reference peak opportunities successfully detected:

$$P_D = \frac{TP}{TP + FN} \tag{28}$$

Expressed as a percentage:

$$P_D(\%) = \frac{TP}{TP + FN} \times 100 \tag{29}$$

5.3 Precision

Precision measures the proportion of detected peaks that correspond to reference peaks:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{30}$$

5.4 Recall

Recall measures the proportion of reference peaks that are successfully detected:

$$\text{Recall} = \frac{TP}{TP + FN} \tag{31}$$

For this experiment, recall is mathematically equal to the aggregate detection probability.

5.5 F1-score

The F1-score is the harmonic mean of precision and recall:

$$F_1 = 2\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{32}$$

The F1-score provides a combined measure of detection completeness and detection reliability.

5.6 False Peaks per Trial

The average number of unmatched detections per trial is:

$$N_{\text{FP,avg}} = \frac{FP}{N_{\text{trials}}} \tag{33}$$

This metric directly quantifies the average number of false peaks generated by the detector in each trial.

5.7 Delay Mean Absolute Error

For M matched peak estimates, the delay MAE is:

$$\text{MAE} = \frac{1}{M} \sum_{i=1}^{M} |\widehat{\tau}_i - \tau_i| \tag{34}$$

The result is expressed in nanoseconds.

5.8 Delay Root Mean Square Error

The delay RMSE is:

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_{i=1}^{M} (\widehat{\tau}_i - \tau_i)^2} \tag{35}$$

RMSE penalizes larger estimation errors more strongly than MAE.

5.9 Delay Bias

The average signed delay estimation error is:

$$\text{Bias} = \frac{1}{M} \sum_{i=1}^{M} (\widehat{\tau}_i - \tau_i) \tag{36}$$

A positive bias indicates average delay overestimation, while a negative bias indicates average delay underestimation.

5.10 Confidence Interval

The detection probability is also evaluated using a 95% confidence interval calculated from the 100 trial-level detection rates.

Let r_t denote the detection rate in trial t.

The mean detection rate is:

$$\overline{r} = \frac{1}{N} \sum_{t=1}^{N} r_t \tag{37}$$

The sample standard deviation is:

$$s_r = \sqrt{\frac{1}{N-1}\sum_{t=1}^{N}(r_t - \overline{r})^2} \tag{38}$$

The standard error is:

$$SE = \frac{s_r}{\sqrt{N}} \tag{39}$$

The approximate 95% confidence interval is:

$$CI_{95\%} = \overline{r} \pm 1.96SE \tag{40}$$

The interval is bounded between 0 and 1 before conversion into percentages. This is a normal-approximation interval based on trial-level detection rates.

6 Experimental Results

6.1 Aggregate Detection Performance

The experiment was conducted at six SNR levels, with 100 Monte Carlo trials per condition.

The aggregate confusion counts and detection metrics are presented in Table 3.

Table 3: Aggregate detection counts for each SNR condition

SNR (dB)	TP	FP	FN	Reference opportunities
-5	422	781	78	500
0	421	461	79	500
5	422	116	78	500
10	455	37	45	500
15	458	3	42	500
20	464	0	36	500

The corresponding detection metrics are presented in Table 4.

Table 4: Refined detection performance under different SNR conditions

SNR (dB)	Detection (%)	Precision (%)	Recall (%)	F1-score	False/trial
-5	84.40	35.08	84.40	0.496	7.81
0	84.20	47.73	84.20	0.609	4.61
5	84.40	78.44	84.40	0.813	1.16
10	91.00	92.48	91.00	0.917	0.37
15	91.60	99.35	91.60	0.953	0.03
20	92.80	100.00	92.80	0.963	0.00

6.2 Mathematical Verification at 20 dB

At 20 dB, the measured aggregate counts were:

$$TP = 464, \quad FP = 0, \quad FN = 36 \tag{41}$$

The detection probability is:

$$P_D = \frac{464}{464 + 36} \tag{42}$$

$$= \frac{464}{500} \tag{43}$$

$$= 0.928 \tag{44}$$

Therefore:

$$\boxed{P_D = 92.80\%} \tag{45}$$

The precision is:

$$\text{Precision} = \frac{464}{464 + 0} \tag{46}$$

$$= 1 \tag{47}$$

Therefore:

$$\boxed{\text{Precision} = 100\%} \tag{48}$$

The recall is:

$$\boxed{\text{Recall} = 92.80\%} \tag{49}$$

The F1-score is:

$$F_1 = 2\frac{1.0 \times 0.928}{1.0 + 0.928} \tag{50}$$

$$\approx 0.9627 \tag{51}$$

Therefore:

$$\boxed{F_1 = 0.963} \tag{52}$$

The average false peaks per trial are:

$$N_{\text{FP,avg}} = \frac{0}{100} = 0 \tag{53}$$

No false peaks were observed in the 100 trials at 20 dB.

6.3 Mathematical Verification at -5 dB

At -5 dB, the measured counts were:

$$TP = 422, \quad FP = 781, \quad FN = 78 \tag{54}$$

The detection probability is:

$$P_D = \frac{422}{422 + 78} \tag{55}$$

$$= \frac{422}{500} \tag{56}$$

$$= 0.844 \tag{57}$$

Therefore:

$$P_D = 84.40\% \tag{58}$$

The precision is:

$$\text{Precision} = \frac{422}{422 + 781} \tag{59}$$

$$\approx 0.3508 \tag{60}$$

Therefore:

$$\text{Precision} = 35.08\% \tag{61}$$

The F1-score is:

$$F_1 = 2\frac{0.3508 \times 0.844}{0.3508 + 0.844} \tag{62}$$

$$\approx 0.4956 \tag{63}$$

The average false peaks per trial are:

$$N_{\text{FP,avg}} = \frac{781}{100} = 7.81 \tag{64}$$

This demonstrates that a high proportion of reference peaks can still be detected at low SNR, while the large number of false detections significantly reduces precision.

7 Delay Estimation Accuracy

The fractional-delay interpolation method was applied to each detected local maximum before matching.

The resulting delay estimation errors are summarized in Table 5.

Table 5: Delay estimation accuracy under different SNR conditions

SNR (dB)	MAE (ns)	RMSE (ns)	Bias (ns)	Matched estimates
-5	2.0971	3.3343	0.5087	422
0	1.7871	2.9914	0.4463	421
5	1.1237	2.0300	-0.0066	422
10	0.6238	1.0246	0.0053	455
15	0.3444	0.5584	-0.0185	458
20	0.2146	0.3538	-0.0023	464

7.1 Example RMSE Calculation at -5 dB

The delay RMSE is calculated using all 422 matched peak estimates:

$$\mathrm{RMSE}_{-5} = \sqrt{\frac{1}{422}\sum_{i=1}^{422}(\widehat{\tau}_i - \tau_i)^2} \tag{65}$$

The measured result is:

$$\boxed{\mathrm{RMSE}_{-5} = 3.3343 \text{ ns}} \tag{66}$$

7.2 Example RMSE Calculation at 20 dB

At 20 dB, 464 matched peak estimates were obtained:

$$\mathrm{RMSE}_{20} = \sqrt{\frac{1}{464}\sum_{i=1}^{464}(\widehat{\tau}_i - \tau_i)^2} \tag{67}$$

The measured result is:

$$\boxed{\mathrm{RMSE}_{20} = 0.3538 \text{ ns}} \tag{68}$$

The reduction in RMSE is:

$$\text{RMSE reduction} = \frac{3.3343 - 0.3538}{3.3343} \times 100 \tag{69}$$

$$\approx 89.39\% \tag{70}$$

Thus, the measured delay RMSE decreases by approximately 89.39% between -5 dB and 20 dB.

This improvement demonstrates the benefit of increased SNR and fractional-sample interpolation under the simulated conditions.

8 Individual Peak Detection Analysis

Aggregate detection probability does not reveal whether all reference peaks have similar detection performance.

Therefore, the detection probability of each of the five reference peaks was evaluated independently.

Table 6 presents the results.

Table 6: Individual reference peak detection probability

Peak	Delay (ns)	-5 dB	5 dB	10 dB	20 dB
1	131.15	100%	100%	100%	100%
2	149.60	100%	100%	100%	100%
3	176.49	69%	71%	95%	100%
4	230.03	69%	59%	63%	64%
5	281.62	84%	92%	97%	100%

The first two peaks are consistently detected across all tested SNR conditions.

Peak 3 exhibits improved detection performance as the SNR increases, reaching 100% at 20 dB.

Peak 5 also improves, reaching 100% at 15 and 20 dB.

However, Peak 4 remains difficult to detect. Its detection probability is only 64% at 20 dB.

This behaviour is associated with its weak reference magnitude, approximately -24.807 dB, which is very close to the -25 dB detection threshold.

The difference between the peak magnitude and the threshold is:

$$\Delta P = (-24.807) - (-25.000) \tag{71}$$

$$= 0.193 \text{ dB} \tag{72}$$

The limited threshold margin explains why the peak detector can miss this component even at relatively high SNR.

The result highlights the importance of selecting a suitable detection threshold for weaker multipath components.

9 Confidence Interval Analysis

The trial-level detection rates were used to calculate approximate 95% confidence intervals.

The resulting intervals are presented in Table 7.

Table 7: 95% confidence intervals for detection probability

SNR (dB)	Detection (%)	Lower bound (%)	Upper bound (%)
-5	84.40	81.42	87.38
0	84.20	81.18	87.22
5	84.40	81.69	87.11
10	91.00	88.96	93.04
15	91.60	89.58	93.62
20	92.80	90.91	94.69

The intervals indicate the variability of the trial-level detection probability across the Monte Carlo simulations.

The confidence intervals at higher SNR are shifted towards higher detection probabilities.

These intervals use a normal approximation and should be interpreted as descriptive estimates for the current simulation setup.

10 Graphical Results

10.1 Detection and Estimation Performance

Figure 1 presents the refined performance metrics across the six SNR conditions.

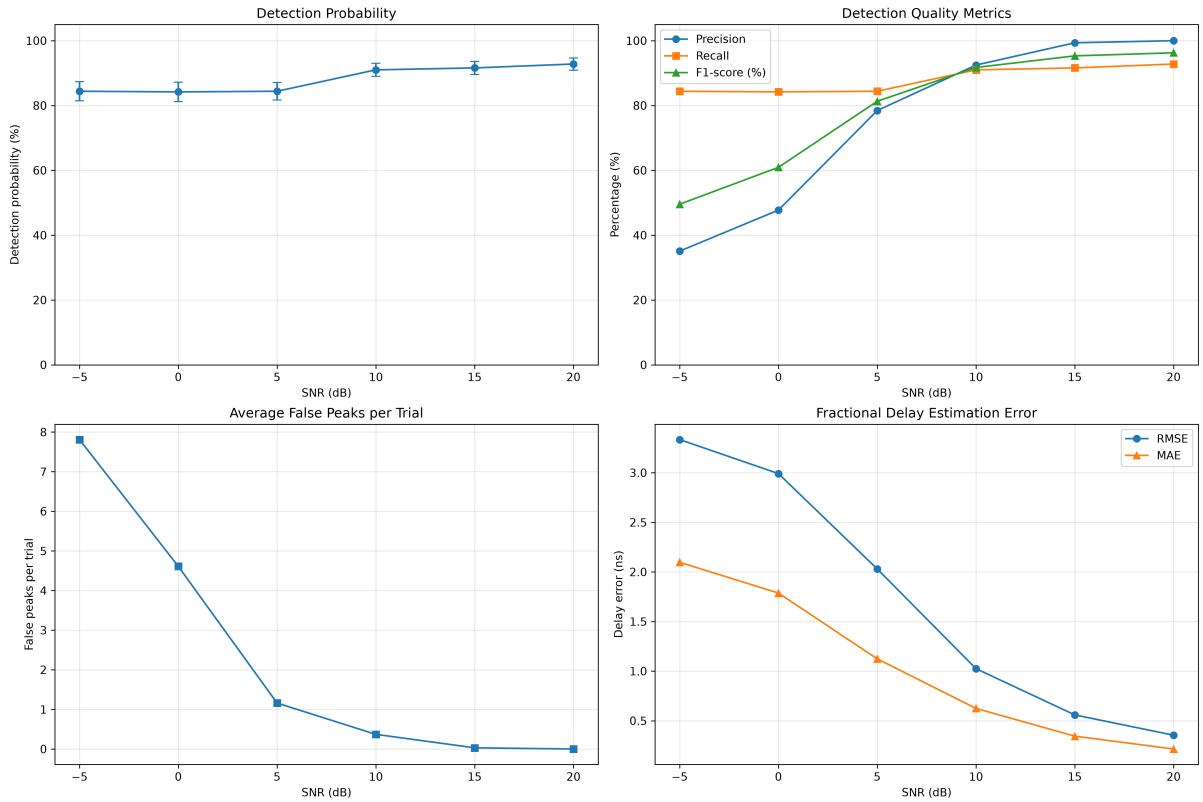


Figure 1: Refined Experiment 15 performance analysis. Top-left: detection probability with approximate 95% confidence intervals. Top-right: precision, recall and F1-score. Bottom-left: average false peaks per trial. Bottom-right: fractional-delay estimation MAE and RMSE.

10.1.1 Detection probability

Detection probability increases from 84.40% at -5 dB to 92.80% at 20 dB.

The trend is not strictly monotonic at the lower SNR conditions. This is expected because the detector operates with a fixed threshold and individual reference peaks have different amplitudes.

10.1.2 Precision, recall and F1-score

Precision increases significantly with SNR.

At -5 dB, precision is 35.08%, indicating that many detected local maxima are not associated with reference peaks.

At 20 dB, precision reaches 100%, with no false detections observed in the 100 trials.

The F1-score increases from 0.496 to 0.963, indicating improved balance between precision and recall.

10.1.3 False peaks

The average false peaks per trial decrease from 7.81 at -5 dB to zero at 20 dB.

This is one of the most significant findings of the experiment.

The reduction demonstrates that noise-induced local maxima become less prominent as the received signal quality improves.

10.1.4 Fractional-delay estimation error

The delay MAE and RMSE both decrease as SNR increases.

The RMSE decreases from 3.3343 ns at -5 dB to 0.3538 ns at 20 dB.

This demonstrates improved stability in the estimated delay positions.

10.2 Delay Profiles at Different SNR Conditions

Figure 2 compares the estimated delay profiles at 0, 10 and 20 dB with the noise-free reference.

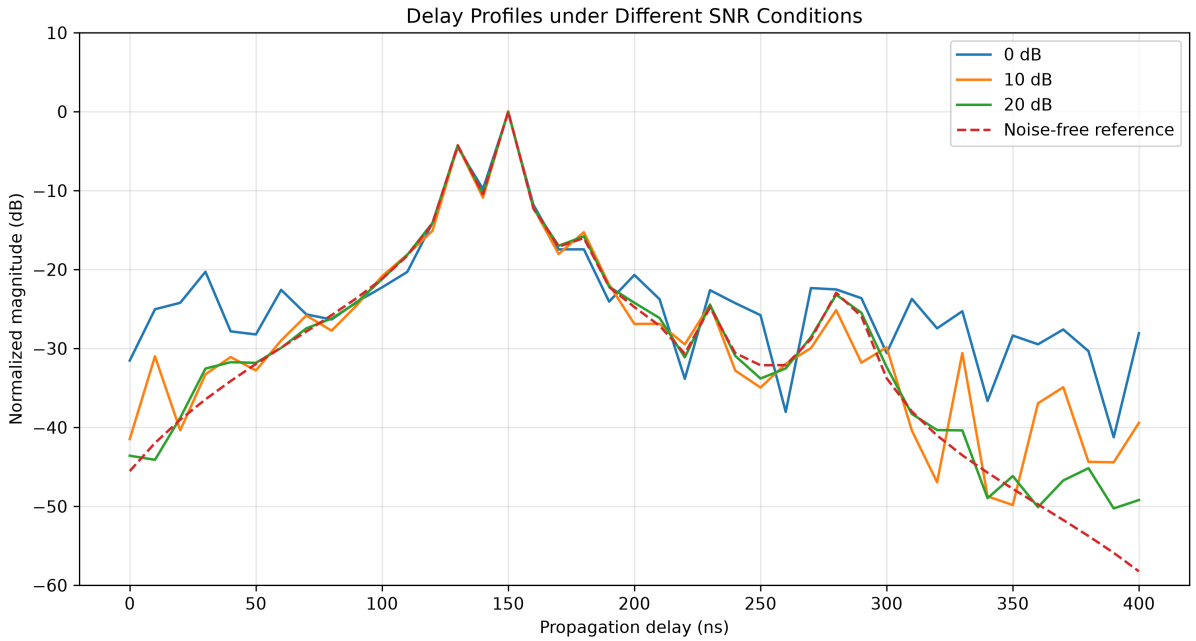


Figure 2: Estimated delay profiles at 0, 10 and 20 dB compared with the noise-free reference delay profile.

At 0 dB, the delay profile exhibits noticeable fluctuations caused by noise.

At 10 dB, the main delay-domain components become more clearly identifiable.

At 20 dB, the estimated profile more closely follows the reference profile, particularly around the dominant components.

The weaker components remain sensitive to the detection threshold and noise.

11 Discussion and Research Findings

The refined evaluation provides several important observations regarding OFDM delay estimation under noisy channel conditions.

11.1 Effect of SNR on Detection Reliability

Increasing SNR substantially improves detection precision.

The measured precision increases from 35.08% at -5 dB to 100% at 20 dB.

This indicates that noise-generated local maxima are progressively suppressed as the signal quality improves.

11.2 Effect of SNR on Delay Accuracy

The delay RMSE decreases from 3.3343 ns to 0.3538 ns across the evaluated SNR range.

The fractional-sample interpolation method allows the estimated peak position to be located between the original 10 ns sampling positions.

The results therefore demonstrate improved sub-sample delay estimation under higher SNR conditions.

11.3 Weak Multipath Detection

The individual peak analysis demonstrates that not all propagation components exhibit the same detection behaviour.

In particular, Peak 4 remains difficult to identify because its reference magnitude is very close to the fixed detection threshold.

This indicates that the detection threshold must be selected according to the relative strength of the expected multipath components and the noise conditions.

11.4 Detection Probability and Precision

The experiment demonstrates that detection probability and precision describe different aspects of detector performance.

At -5 dB, the detection probability is 84.40%, but precision is only 35.08%.

This means that many reference peaks are detected, but a large number of additional noise-generated peaks are also reported.

At 20 dB, precision reaches 100%, while detection probability remains 92.80%.

Therefore, no false detections were observed, but some reference peaks were still missed.

11.5 Effect of Fractional-Delay Interpolation

The parabolic interpolation method reduces quantization of the estimated peak position to integer sample locations.

The resulting delay estimates are not restricted to multiples of 10 ns.

However, fractional-sample interpolation does not guarantee sub-nanosecond physical resolution. Its accuracy depends on the sampled peak shape, SNR, effective bandwidth and interpolation model.

12 Limitations

The following limitations apply to the experiment:

1. **Static propagation channel:** The simulation does not include time-varying propagation or Doppler effects caused by vehicle movement.
2. **Ideal synchronization:** Noise is added after ideal cyclic-prefix removal and synchronization. Timing and carrier-frequency synchronization errors are not evaluated.
3. **Single OFDM symbol:** The evaluation uses one known QPSK OFDM symbol for each trial.
4. **Fixed detection threshold:** The -25 dB threshold is not adaptive. Weak components close to the threshold may be missed.
5. **Finite bandwidth:** The 100 MHz bandwidth limits the ability to separate closely spaced propagation components.
6. **Reference peak definition:** The five reference peaks are algorithmically extracted from the noise-free discrete channel. They do not necessarily correspond one-to-one with individual physical rays.
7. **Monte Carlo sample size:** Each SNR condition uses 100 trials. A larger number of trials would provide more extensive statistical evaluation.
8. **Confidence interval approximation:** The reported intervals use a normal approximation of trial-level detection rates.
9. **No hardware validation:** The results are simulation-based and have not yet been validated using USRP hardware.

13 Relevance to V2X-ISAC Research

The experiment provides a simulation baseline for evaluating multipath delay estimation in V2X Integrated Sensing and Communication systems.

Delay estimation is relevant to sensing applications because propagation delays provide information about the geometry of the transmitter, receiver and surrounding propagation environment.

For a conventional monostatic radar, the nominal range-resolution scale is:

$$\Delta R \sim \frac{c}{2B} \tag{73}$$

For an illustrative bandwidth of 100 MHz:

$$\Delta R \sim \frac{299792458}{2(100 \times 10^6)} \tag{74}$$

$$\approx 1.50 \text{ m} \tag{75}$$

This is a theoretical monostatic bandwidth-based range-resolution estimate, not a measured result of the current experiment.

The current experiment uses spatially separated transmitter and receiver locations. Consequently, the propagation delays must be interpreted using bistatic geometry rather than directly applying the monostatic range equation.

The improved delay estimation results provide a basis for subsequent investigations involving:

- Adaptive multipath peak detection.
- Weak target and reflection detection.
- Higher-bandwidth OFDM waveforms.
- Delay-Doppler processing.
- Bistatic localization.
- Hardware-based validation using SDR platforms.

14 Conclusion

Experiment 15 evaluated the robustness of OFDM-based delay estimation under six different SNR conditions using a multipath channel derived from Sionna RT at 5.9 GHz.

A refined peak detection methodology was implemented using one-to-one peak matching and parabolic fractional-sample interpolation.

The evaluation demonstrated that increasing SNR improves detection reliability and delay estimation accuracy.

Precision increased from 35.08% at -5 dB to 100% at 20 dB, while the F1-score increased from 0.496 to 0.963.

The average false peaks decreased from 7.81 per trial to zero, and delay RMSE decreased from 3.3343 ns to 0.3538 ns.

The individual peak analysis also identified a weaker delay component that remained difficult to detect because its magnitude was close to the selected detection threshold.

These results demonstrate that multipath delay estimation performance depends on both SNR and the relative strength of individual propagation components.

The experiment establishes a more rigorous simulation baseline for subsequent V2X-ISAC research, particularly for adaptive delay estimation and weak multipath component detection.